

Hoff Bayesian Statistics

Contents

1	Chapters	1
2	Final Project	1

These are (fully reproducible!) R Markdown lecture notes for Hoff (2009), completed as part of a 1-semester independent study course. Only Chapters 1-8 are complete right now.

Each note includes summaries of chapter sections, with math and explanations modified to better fit my understanding and the occasional link to external resources. I also reproduce many figures in the book in a ggplot/tidyverse style, and tackle some of the exercises at the end of each chapter (correctness not guaranteed).

If you find an error or would like to improve the notes, please let me know/submit a PR!

1 Chapters

- Chapter 1: Introduction and examples¹
- Chapter 2: Belief, probability, and exchangeability²
- Chapter 3: One-parameter models³
- Chapter 4: Monte Carlo approximation⁴
- Chapter 5: The Normal Model⁵
- Chapter 6: Posterior approximation with the Gibbs sampler⁶
- Chapter 7: The multivariate normal model⁷
- Chapter 8: Group comparisons and hierarchical modeling⁸
- Chapter 9: Linear regression⁹
- Chapter 10: Nonconjugate priors and Metropolis-Hastings algorithms¹⁰

2 Final Project

As a small final project, I also implemented R code for the basic binary relation version of the Infinite Relational Model, described in Kemp et al. (2006).

- Infinite Relational Model¹¹

Hoff, Peter D. 2009. *A First Course in Bayesian Statistical Methods*. Springer Texts in Statistics. Springer. <https://doi.org/10.1007/978-0-387-92407-6>.

¹1.qmd

²2.qmd

³3.qmd

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¹¹irm.qmd

Kemp, Charles, Joshua B. Tenenbaum, Thomas L. Griffiths, Takeshi Yamada, and Naonori Ueda. 2006. “Learning Systems of Concepts with an Infinite Relational Model.” *Proceedings of the National Conference on Artificial Intelligence (AAAI)* 21: 381–88. <http://web.mit.edu/cocosci/Papers/Kemp-et al-AAAI06.pdf>.